

[< Corn](#)**AGRICULTURE**

Corn Earworm

Helicoverpa zea

Updated: 08/2008



On this page



Description of the Pest

[Corn earworm moths](#) are most active during evening and night.

They are about 0.75 inch long, rather robust, with a wing span of 1 to 1.5 inches, and adults range from olive green, to tan, to dark reddish brown in color. [Egg](#) laying occurs throughout the sweet corn growing season. The tiny, white eggs are laid singly on the foliage and fresh corn silk, which is the favorite site for egg deposition. After about a day, eggs develop a reddish brown ring in the top portion. Eggs are

spherical with 12 or more ridges radiating from the top. [Young larvae](#) are greenish with black heads and conspicuous black hairs on the body. [Fully developed worms](#) are about 1.5 inches long and range in color from pale green or pinkish to brown.

Damage

The corn earworm may be present throughout the season but is most abundant during August and September. Larvae feed on leaves, tassels, the whorl, and within ears, but the ears are the preferred sites for corn earworm attack. [Ear damage](#) is characterized by extensive excrement at the ear tip. Young larvae feed on corn silks, clipping them off. Shortly thereafter, they feed their way into the ear where they remain, feeding in the tip area until they exit to [pupate](#) in the soil.

Management

Corn earworm is primarily a problem in sweet corn where treatments should be timed to coincide with egg hatch.

Biological Control

Many predators and parasites attack corn earworm eggs, including several species of [Trichogramma](#). Most [parasitized eggs](#) turn black, but there may be a lag period before they do so. General predators such as [lacewings](#), [minute pirate bugs](#), and [damselflies](#) feed on corn earworm eggs and small larvae.

Cultural Control

In sweet corn, very early plantings require fewer treatments than late-season corn because earworm population densities increase as the season progresses.

Organically Acceptable Methods

Biological and cultural controls and sprays of *Bacillus thuringiensis* and the Entrust formulation of spinosad are acceptable for use on an organically grown crop.

Monitoring and Treatment Decisions

Insecticidal control of corn earworm is difficult and depends on proper timing and thorough coverage. Begin sampling soon after corn emergence but pay particular attention to corn that is silking in late summer/early fall. The presence of large numbers of eggs on fresh corn silks indicates the potential for damaging populations. Eggs hatch in 5 to 7 days following oviposition. Once larvae enter the corn ears, control with insecticides is difficult. Direct insecticidal control towards young larvae that are feeding on the exposed ear tips. Treatments are usually not needed on field or silage corn. In sweet corn, where tolerance for worm damage is low, timing of insecticide treatments is critical: begin treatments during silking stage, at the start of egg hatch. Apply additional treatments if they are necessary.

Pesticides and Natural Enemy Releases

Not all registered pesticides are listed. **The following are ranked by their IPM value, with the most effective and least harmful to natural enemies, honey bees, and the [environment](#) listed at the top of the table.** When choosing a pesticide, consider information related to water and [air quality](#), resistance management, and the pesticide's properties and application timing. Use [PestManage](#) to compare summarized management options for different pests in the same crop. Always carefully read the label of the product being used and take all necessary [precautions when handling pesticides](#).



 [Expand table](#)

Rank 	Active ingredient 	Example 1
Application Timing Varies (See <i>UC IPM Pest Management</i>		
A	spinosad	Entrust 
A	spinosad	Success
B	spinetoram	Radiant SC
C	methomyl 	Lannate S
D	esfenvalerate	Asana XL
E	permethrin	Pounce 3.
F	Bacillus thuringiensis ssp. kurstaki	various pr

I Do not apply or allow to drift to plants that are flowering including weeds. Do not allow pesticide to contaminate water accessible to bees including puddles.

II Do not apply or allow to drift to plants that are flowering including weeds, except when the application is made between sunset and midnight if allowed by the pesticide label and regulations. Do not allow pesticide to contaminate water accessible to bees including puddles.



No bee precaution, except when required by the pesticide label or regulations.

- ‡ [a b](#) Restricted entry interval (REI) is the number of hours (unless otherwise noted) from treatment until the treated area can be safely entered without personal protective equipment. Preharvest interval (PHI) is the number of days from treatment to harvest. In some cases, the REI exceeds the PHI. The longer of the two intervals is the minimum time that must elapse before harvest.
- * [a](#) Permit required from county agricultural commissioner for purchase or use.
- # [a b](#) Acceptable for use on certified organic crops. Check with your certifier to confirm before application.
- ¹ [↩](#) Group numbers for insecticides and miticides are assigned by the [Insecticide Resistance Action Committee](#) (IRAC). Insecticides with unknown modes of action are assigned mode-of-action group numbers (MoAs) that begin with UN. Rotate pesticides with a different mode-of-action group number, and do not use products with the same mode-of-action group number more than twice per season to help prevent the development of resistance. For example, the organophosphates have a group number of 1B; insecticides with a 1B group number should be alternated with insecticides that have a group number other than 1B.
- ² [↩](#) Range of insect and mite groups affected by a pesticide. **Broad** means the pesticide affects most groups of insects and mites; **narrow** means the pesticide affects only a few specific groups.
- ³ [↩](#) Risk of harm to honey bees. For more information, see [Bee Precaution Pesticide Ratings](#).
- ⁴ [↩](#) Risk of harm to predatory mites. Toxicities are generally to western predatory mite, *Galendromus occidentalis*. Where differences have been measured in toxicity of the pesticide-

resistant strain versus the native strain, these are listed as pesticide-resistant strain or native strain.

- ⁵ [a b](#) Risk of harm to parasitoids and general predators. Toxicities are averages of reported effects and should be used only as a general guide. Actual toxicity of a specific insecticide depends on factors including the application rate, environmental conditions, and the life stage and species of a parasitoid or predator.
- ⁶ [↩](#) Length of time residue affects natural enemies. **Short** means hours to days; **moderate** means days to 2 weeks; and **long** means many weeks or months.
- ⁷ [↩](#) Risk of harm to fish from leaching, based on USDA Natural Resources Conservation Service Windows Pesticide Screening Tool (WIN-PST).
- ⁸ [↩](#) Risk of harm to fish from adsorbed runoff, based on USDA Natural Resources Conservation Service Windows Pesticide Screening Tool (WIN-PST).
- ⁹ [↩](#) Risk of harm to fish from solution runoff, based on USDA Natural Resources Conservation Service Windows Pesticide Screening Tool (WIN-PST).
- ¹⁰ [↩](#) Risk of harm to humans from leaching, based on USDA Natural Resources Conservation Service Windows Pesticide Screening Tool (WIN-PST).
- ¹¹ [↩](#) Risk of harm to humans from solution runoff, based on USDA Natural Resources Conservation Service Windows Pesticide Screening Tool (WIN-PST).
- ¹² [↩](#) Date information was last updated in the *UC IPM Pest Management Guidelines*.
- ¹³ [a b c d e](#) Toxic to some natural enemies (lacewing and syrphid fly larvae, predatory beetles, and thrips) when sprayed and up to 5 to 7 days after.
- ¹⁴ [↩](#) Residual is moderate if solution is between a pH of 7 and 8.



UC IPM Pest Management Guidelines: Corn

UC ANR Publication 3443

L.D. Godfrey, Entomology, UC Davis

[S.D. Wright](#) (emeritus), UC Cooperative Extension Tulare and Kings counties

[C.G. Summers](#) (emeritus), Entomology, UC Davis and Kearney Agricultural Research and Extension Center, Parlier

C.A. Frate (emeritus), UC Cooperative Extension Tulare County
Acknowledgement for Contributions to Invertebrates

[M.J. Jimenez](#) (emeritus), UC Cooperative Extension Tulare County

Managing Pests

About

Agriculture

About UC IPM

Home and Landscape

Annual Reports

Natural Area Pests

UC IPM Strategic Plan

Pesticide Safety

UC IPM Programs

Invasive and Exotic Pests

What is IPM?

What's New

Resources

Get In Touch

Glossary

Contact Us

[Links](#)

[Feedback](#)

[Pesticide Information](#)

[Publications and Newsletter](#)

[Recursos en español](#)

[Trainings and Events](#)



UNIVERSITY OF CALIFORNIA
Agriculture and Natural Resources

Statewide Integrated
Pest Management Program



Home and Landscape: [Instagram](#) | [Facebook](#)

[Use Permissions and Citations](#) | [Nondiscrimination Statement](#) |
[Acknowledgements](#) | [Accessibility](#) | [Privacy Statement](#) | [Staff-Only Page](#)

©1996—2026 Statewide IPM Program, Agriculture and Natural Resources,
University of California Regents of the University of California unless otherwise
noted.